

Rail risk register | management review

Risk method

SIMULATED register: six risks; expected monetary exposure GBP 14.30m. Exposure=probability*impact; ordinal score=ceil(probability*5)*impact band, with red>=15, amber>=8. Probability/impact are scenario assumptions. Named accountable roles, mitigations and due dates are in risks.csv.

Priority mitigation

Prioritize S3 possession access and electrification supply review. Its simulated lower productivity and higher cost factor explain weak delivery; this is a designed scenario, not a causal finding from ORR. Confirm access, design and supplier readiness before committing recovery spend. Monitor physical units, weekly earned value, unpaid commitments, overdue actions and forecast movement.

Risk-by-risk action register

R01 | S3 | Possession access restricted. Probability 65%; impact GBP 8.00m; expected exposure GBP 5.20m; RED, ordinal score 20. Owner: Construction manager. Due: 2026-08-17. Mitigation: Agree alternative possession windows. Status: open, no closure evidence assumed.

R02 | S3 | Electrification supply lead time. Probability 55%; impact GBP 5.00m; expected exposure GBP 2.75m; AMBER, ordinal score 12. Owner: Supply chain manager. Due: 2026-08-24. Mitigation: Reserve approved alternate supplier. Status: open, no closure evidence assumed.

R03 | S2 | Station interface rework. Probability 35%; impact GBP 3.00m; expected exposure GBP 1.05m; GREEN, ordinal score 6. Owner: Design manager. Due: 2026-08-31. Mitigation: Close design interface reviews. Status: open, no closure evidence assumed.

R04 | ALL | Safety assurance approval delay. Probability 30%; impact GBP 7.00m; expected exposure GBP 2.10m; AMBER, ordinal score 10. Owner: Assurance manager. Due: 2026-09-07. Mitigation: Independent assurance readiness review. Status: open, no closure evidence assumed.

R05 | S4 | Ground conditions at station. Probability 20%; impact GBP 4.00m; expected exposure GBP 0.80m; GREEN, ordinal score 4. Owner: Geotechnical lead. Due: 2026-08-10. Mitigation: Targeted ground investigation. Status: open, no closure evidence assumed.

R06 | ALL | Signaling integration test failure. Probability 40%; impact GBP 6.00m; expected exposure GBP 2.40m; AMBER, ordinal score 8. Owner: Systems manager. Due: 2026-09-14. Mitigation: Early test rig integration. Status: open, no closure evidence assumed.

Approval and monitoring

Construction manager owns R01, supply chain manager owns R02. Review weekly; escalation requires overdue mitigation or score>=15. Validate recovery feasibility and approve a formal baseline change before substituting any

recovery dates.

Limitations

Calendar forecasts extrapolate each package's last four intervals independently; this is not a resource-leveled critical-path schedule, and no statistical confidence interval is claimed. Budget-weighted physical progress is not average miles. SPI becomes uninformative at final completion. ORR four-week periods are not months; moving annual values and missing flags are preserved. Route-km and track-km are distinct; infrastructure breaks preclude naive change claims.

Source boundary

OBSERVED CONTEXT: ORR table 3181a supplies 96 railway periods of regional delay rates; table 6320 supplies 79 annual nation records. Context is not used to calibrate simulated project progress. ORR data is licensed OGL v3.0; source Network Rail (infrastructure also TfL and Amey Infrastructure Wales). Source URLs and SHA256 hashes: data/manifest.json.

